

Analog Circuits Tutorial: Solutions

Theory Pointers and Worked Numericals

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Solutions correspond to Q1–Q30 of the accompanying tutorial.

Unless specified otherwise: $V_{BE} = 0.7\text{ V}$, $V_{CE(\text{sat})} = 0.2\text{ V}$, $V_T = 25\text{ mV}$, and the Early effect is neglected. Numerical values are approximate.

Part A: Theory and viva pointers

Q1. BJT basics

- Emitter: heavily doped to inject carriers efficiently.
- Base: thin and lightly doped, minimizing recombination.
- Collector: designed to collect carriers and withstand voltage and power dissipation.
- $I_E = I_C + I_B$.
- $\alpha = I_C/I_E$, $\beta = I_C/I_B$.
- $\beta = \frac{\alpha}{1 - \alpha}$, $\alpha = \frac{\beta}{\beta + 1}$.

Q2. Operating regions

- Cutoff: negligible forward base-emitter drive; collector current approximately zero.
- Forward-active: BE junction forward biased, BC junction reverse biased; suitable for linear amplification.
- Saturation: both junctions forward biased.
- In saturation, external circuit resistances largely limit collector current; $I_C = \beta I_B$ is no longer reliable.

Q3. Transistor voltages

- V_{BE} depends on current, temperature, and device.
- $V_{CE(\text{sat})}$ depends on collector current, base drive, temperature, and device.
- $V_{CE} = V_{BE} - V_{BC}$.
- In saturation, both junction voltages are positive; their difference can be small, for example $0.8 - 0.6 = 0.2\text{ V}$.

Q4. Bias, Q-point, and swing

- Bias establishes steady currents and voltages.
- Q-point is the operating point with the input signal zero.
- For $i_C = I_{CQ} + I_m \sin \omega t$, peak AC amplitude is I_m , and peak-to-peak variation is $2I_m$.
- Bias supplies the nonzero resting collector current.
- A Q-point near an operating limit leaves little room for one half-cycle and causes early clipping.

Q5. DC and AC load lines

- DC: $V_{CC} = I_C R_C + V_{CE} + I_E R_E$.
- With $I_E \approx I_C$: $V_{CE} \approx V_{CC} - I_C(R_C + R_E)$.
- Coupling capacitors block DC but connect signal loads.
- Bypass capacitors retain DC emitter feedback but can remove AC emitter feedback.
- AC-grounded emitter: $\Delta v_{CE} = -(R_C \parallel R_L) \Delta i_C$.
- Unbypassed emitter: $\Delta v_{CE} \approx -[(R_C \parallel R_L) + R_E] \Delta i_C$.
- Zero signal change corresponds to the Q-point; hence the AC line passes through it.

Q6. Bias circuits

- Fixed bias: $I_B = (V_{CC} - V_{BE})/R_B$, $I_C \approx \beta I_B$.
- Divider: $V_{TH} = V_{CC} R_2 / (R_1 + R_2)$, $R_{TH} = R_1 \parallel R_2$.

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1)R_E}.$$

- Base current lowers actual base voltage below the unloaded divider voltage.
- R_E reduces sensitivity to transistor variation.

Q7. Temperature and stability

- Increased current raises V_E , reducing V_{BE} if V_B is approximately fixed.
- At fixed current, required V_{BE} generally decreases with increasing temperature; leakage also increases.
- Thermal runaway: heating increases current and dissipation, causing further heating.
- Countermeasures: stable bias, emitter feedback, appropriate heat sinking, and operation within device ratings.

Q8. Large- and small-signal models

- $I_C = I_S e^{V_{BE}/V_T}$.
- Linearize at Q: $i_c = (\partial I_C / \partial V_{BE})_Q v_{be} = g_m v_{be}$.
- $g_m = I_{CQ} / V_T$; bias current sets the local slope.

- Valid for sufficiently small variations in forward-active operation; does not model clipping accurately.

Q9. Hybrid- π intuition

- r_π : incremental base-current response, $i_b = v_{be}/r_\pi$.
- Controlled source: collector-current response $i_c = g_m v_{be}$.
- r_o : collector-current dependence on v_{ce} due to the Early effect.
- Since $i_c = \beta i_b$, $r_\pi = \beta/g_m$.
- r_π is a differential resistance, not the ratio of DC voltage to DC current.

Q10. CE gain

- Let $R'_C = R_C \parallel R_L$.
- AC-grounded emitter: $A_v = -g_m R'_C$.
- Unbypassed emitter:

$$A_v = -\frac{\beta R'_C}{r_\pi + (\beta + 1)R_E}.$$

- More collector current produces less collector voltage: inversion.
- Emitter degeneration lowers gain, raises input resistance, and improves linearity and gain stability.

Q11. Input and output resistance

- Resistance at a port: small test voltage divided by resulting test current.
- For output resistance, remove the load and set independent signal sources to zero; preserve their internal resistances.
- Controlled sources remain because they describe transistor response to the test signal.
- Source loading:

$$\frac{v_o}{v_s} = A_v \frac{R_{in}}{R_S + R_{in}}.$$

Q12. Common base

- Base AC-grounded: $v_{be} = -v_e$.
- $A_v = +g_m(R_C \parallel R_L)$.
- $R_{in} = r_\pi/(\beta + 1) \approx 1/g_m$.
- Increasing emitter voltage reduces collector current, increasing collector voltage: noninverting.
- Collector-to-emitter signal-current gain magnitude is approximately $\alpha < 1$; voltage gain can still be large.

Q13. Emitter follower

- With $R'_E = R_E \parallel R_L$:

$$A_v = \frac{(\beta + 1)R'_E}{r_\pi + (\beta + 1)R'_E}.$$

- $R_{\text{in,base}} = r_\pi + (\beta + 1)R'_E$.
- Positive gain slightly below unity.
- Useful for buffering: high input resistance and low output resistance.
- With load removed:

$$R_{\text{out}} \approx R_E \parallel \frac{r_\pi + (R_S \parallel R_B)}{\beta + 1}.$$

Q14. Configuration selection

- Voltage gain: CE, often followed by a buffer if the final load is low.
- High-resistance source to low-resistance load: CC.
- Low-resistance source with voltage gain: CB can be suitable.
- CE inverts; CB and CC do not.
- Assess source attenuation and load resistance before comparing gain.

Q15. Cascade

- Output of one stage drives the next stage's input.
- Stage 1 is loaded by stage 2's input resistance.
- Overall gain is the product of loaded stage gains and initial source attenuation.
- Two CE inversions give a noninverting overall midband output.
- Multiple stages also interact through frequency response and available signal swing.

Q16. Cascode

- Cascade: sequential amplifier stages.
- Cascode: CE input transistor feeding a CB transistor.
- Upper CB transistor holds the lower collector at a relatively small AC voltage variation.
- Reduced lower-transistor BC voltage swing reduces Miller multiplication.
- Often provides wider bandwidth and high transistor-stack output resistance.
- Requires sufficient DC voltage across both devices; headroom and output swing must be checked.

Q17. Low-frequency response

- Input/output coupling capacitors attenuate sufficiently low-frequency signals.
- A bypass capacitor becomes less effective at low frequency, increasing emitter feedback and reducing gain.

- Estimate each corner from $f_c = 1/(2\pi R_{\text{seen}}C)$.
- Set independent sources to zero and use the appropriate termination to find R_{seen} .
- The largest corner is useful as a dominant-pole estimate only if the other effects are sufficiently separated.
- Closely spaced corners require combined response analysis.

Q18. High-frequency response

- C_π , C_μ , and stray/load capacitances cause high-frequency attenuation.
- Current through BC capacitance: $i = C_\mu d(v_b - v_c)/dt$.
- With $v_c = A_v v_b$, input equivalent capacitance is $C_\mu(1 - A_v)$.
- Inverting CE gain gives substantial Miller multiplication.
- At a -3 dB cutoff, voltage gain magnitude is $1/\sqrt{2}$ of midband gain.
- Bandwidth: $f_H - f_L$.
- Measure with a small sinusoidal input while sweeping frequency; monitor the actual input voltage and avoid clipping.

Q19. Class A

- $P_{\text{DC}} = V_{CC}I_{CQ}$; sinusoidal $P_{\text{AC}} = V_m I_m / 2$.
- Series-fed ideal midpoint: $V_m = V_{CC}/2$, $I_m = I_{CQ}$, giving $\eta_{\text{max}} = 25\%$.
- Ideal transformer coupling: $V_m = V_{CC}$, $I_m = I_{CQ}$, giving $\eta_{\text{max}} = 50\%$.
- $I_m \leq I_{CQ}$ avoids negative collector current; saturation can impose an additional limit.
- Transformer voltage permits collector excursions above the supply while its quiescent value remains near V_{CC} .
- Nonzero I_{CQ} means nonzero idle power.

Q20. Class B/AB and distortion

- Class B: each transistor ideally conducts for 180° .
- Push-pull devices handle opposite signal half-cycles.
- For dual rails of magnitude V_{CC} :

$$P_{\text{DC}} = \frac{2V_{CC}I_m}{\pi}, \quad P_o = \frac{V_m I_m}{2}, \quad \eta = \frac{\pi V_m}{4V_{CC}}.$$

- At $V_m = V_{CC}$, $\eta_{\text{max}} = \pi/4 = 78.5\%$.
- Crossover notch: insufficient conduction near zero; Class AB introduces a small idle current.
- Clipping: flattened peaks from swing limits, excessive input, incorrect bias, or supply limitations.
- Diagnose by checking DC bias, supply voltages, input amplitude, and where distortion occurs on the waveform.

Part B: Worked numerical solutions

Q21. Fixed bias and maximum swing

$$I_B = \frac{12 - 0.7}{200 \text{ k}\Omega} = 56.5 \mu\text{A}.$$

For $\beta = 100$,

$$I_C = 5.65 \text{ mA}, \quad V_{CE} = 12 - 5.65 = 6.35 \text{ V}.$$

The collector is above the base and the transistor is forward-active.

Load-line endpoints:

$$\text{Cutoff: } (V_{CE}, I_C) = (12 \text{ V}, 0),$$

$$\text{Saturation: } I_C = \frac{12 - 0.2}{1 \text{ k}\Omega} = 11.8 \text{ mA}.$$

Available collector-voltage excursions:

$$12 - 6.35 = 5.65 \text{ V} \quad \text{toward cutoff,}$$

$$6.35 - 0.2 = 6.15 \text{ V} \quad \text{toward saturation.}$$

Therefore the load-line estimate of maximum symmetric swing is

$$\boxed{V_m = 5.65 \text{ V}, \quad V_{pp} = 11.3 \text{ V}.}$$

For $\beta = 150$,

$$I_C = 8.475 \text{ mA}, \quad V_{CE} = 3.525 \text{ V}.$$

The available downward excursion is only $3.525 - 0.2 = 3.325 \text{ V}$. Saturation now limits the symmetric swing.

Q22. Voltage-divider bias

$$V_{TH} = 12 \frac{12}{39 + 12} = 2.8235 \text{ V},$$

$$R_{TH} = 39 \parallel 12 = 9.1765 \text{ k}\Omega.$$

For $\beta = 100$,

$$I_B = \frac{2.8235 - 0.7}{9.1765 \text{ k}\Omega + 101(470 \Omega)} = 37.49 \mu\text{A}.$$

Hence

$$I_C = 3.749 \text{ mA}, \quad I_E = 3.786 \text{ mA},$$

$$V_E = 1.780 \text{ V}, \quad V_B = 2.480 \text{ V},$$

$$V_C = 8.251 \text{ V}, \quad \boxed{V_{CE} = 6.472 \text{ V}.}$$

The divider voltage falls by approximately $2.8235 - 2.480 = 0.344 \text{ V}$ due to base-current loading.

For $\beta = 150$,

$$I_B = \frac{2.1235}{9.1765 \text{ k}\Omega + 151(470 \Omega)} \approx 26.49 \mu\text{A},$$

$$I_C \approx 3.973 \text{ mA}.$$

The collector-current increase is approximately

$$\frac{3.973 - 3.749}{3.749} \times 100\% \approx \boxed{6.0\%}.$$

A forward-active fixed-bias circuit would show approximately a 50% increase for the same change in β .

Q23. Bias-network design

Required currents:

$$I_B = \frac{2 \text{ mA}}{100} = 20 \mu\text{A}, \quad I_E = 2.02 \text{ mA}.$$

Emitter resistor:

$$R_E = \frac{1.2 \text{ V}}{2.02 \text{ mA}} = \boxed{594.1 \Omega}.$$

Required collector voltage:

$$V_C = V_{CE} + V_E = 7.2 \text{ V}.$$

Therefore

$$R_C = \frac{12 - 7.2}{2 \text{ mA}} = \boxed{2.4 \text{ k}\Omega}.$$

Required base voltage is $V_B = 1.2 + 0.7 = 1.9 \text{ V}$. Set $I_{R_2} = 10I_B = 200 \mu\text{A}$:

$$R_2 = \frac{1.9}{200 \mu\text{A}} = \boxed{9.5 \text{ k}\Omega}.$$

Since $I_{R_1} = I_{R_2} + I_B = 220 \mu\text{A}$,

$$R_1 = \frac{12 - 1.9}{220 \mu\text{A}} = \boxed{45.91 \text{ k}\Omega}.$$

The Thevenin values are approximately

$$V_{TH} = 2.0574 \text{ V}, \quad R_{TH} = 7.8717 \text{ k}\Omega.$$

Verification:

$$I_B = \frac{2.0574 - 0.7}{7.8717 \text{ k}\Omega + 101(594.1 \Omega)} = 20 \mu\text{A}.$$

One nearby E12 choice is

$$R_1 = 47 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega, \quad R_E = 560 \Omega, \quad R_C = 2.2 \text{ k}\Omega.$$

These give

$$\begin{aligned} V_{TH} &= 2.1053 \text{ V}, & R_{TH} &= 8.2456 \text{ k}\Omega, \\ I_B &\approx 21.68 \mu\text{A}, & I_C &\approx 2.168 \text{ mA}, & V_E &\approx 1.226 \text{ V}, \\ V_C &\approx 7.229 \text{ V}, & & & & \boxed{V_{CE} \approx 6.003 \text{ V}.} \end{aligned}$$

Relative to the targets: I_C is approximately 8.4% high, V_E is 2.2% high, and V_{CE} is about 0.06% high. Other sensible E12 choices are acceptable if verified.

Q24. CE small-signal analysis

$$\begin{aligned} g_m &= \frac{2 \text{ mA}}{25 \text{ mV}} = 0.08 \text{ S}, \\ r_\pi &= \frac{100}{0.08} = 1.25 \text{ k}\Omega, & r_o &= \frac{100 \text{ V}}{2 \text{ mA}} = 50 \text{ k}\Omega. \end{aligned}$$

Bypassed emitter:

$$R_C \parallel R_L \parallel r_o = 3.3 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel 50 \text{ k}\Omega \approx 2.3639 \text{ k}\Omega.$$

$$A_{v,b} \approx -0.08(2363.9) = -189.1.$$

Total input resistance:

$$R_{\text{in}} = 20 \text{ k}\Omega \parallel 1.25 \text{ k}\Omega = 1.1765 \text{ k}\Omega.$$

Source-to-output gain:

$$A_{v,s} = -189.1 \frac{1.1765}{1 + 1.1765} \approx \boxed{-102.2}.$$

With R_L removed:

$$R_{\text{out}} = 3.3 \text{ k}\Omega \parallel 50 \text{ k}\Omega \approx \boxed{3.096 \text{ k}\Omega}.$$

Unbypassed emitter, neglecting r_o :

$$R'_C = 3.3 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 2.4812 \text{ k}\Omega,$$

$$R_{\text{in,base}} = 1250 + 101(470) = 48.72 \text{ k}\Omega.$$

$$A_{v,b} = -\frac{100(2481.2)}{48720} \approx \boxed{-5.09},$$

$$R_{\text{in}} = 20 \text{ k}\Omega \parallel 48.72 \text{ k}\Omega \approx \boxed{14.18 \text{ k}\Omega}.$$

Emitter feedback substantially reduces gain and increases input resistance.

Q25. Emitter follower

$$g_m = 0.08 \text{ S}, \quad r_\pi = 1.25 \text{ k}\Omega, \quad R'_E = 1 \text{ k}\Omega \parallel 1 \text{ k}\Omega = 500 \Omega.$$

Base-to-output gain:

$$A_{v,b} = \frac{101(500)}{1250 + 101(500)} = \boxed{0.97585}.$$

$$R_{\text{in,base}} = 51.75 \text{ k}\Omega,$$

$$R_{\text{in}} = 50 \text{ k}\Omega \parallel 51.75 \text{ k}\Omega \approx 25.430 \text{ k}\Omega.$$

Overall gain:

$$A_{v,s} = 0.97585 \frac{25.430}{2 + 25.430} \approx \boxed{0.9047}.$$

For output resistance, remove R_L :

$$R_G = 2 \text{ k}\Omega \parallel 50 \text{ k}\Omega = 1.9231 \text{ k}\Omega,$$

$$R_{\text{out}} = 1 \text{ k}\Omega \parallel \frac{1250 + 1923.1}{101} \approx \boxed{30.46 \Omega}.$$

The follower presents a relatively high input resistance while providing a much lower output resistance for load driving.

Q26. CB and cascode

$$g_m = \frac{1 \text{ mA}}{25 \text{ mV}} = 0.04 \text{ S}, \quad r_\pi = 2.5 \text{ k}\Omega.$$

$$R'_C = 4.7 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 3.1973 \text{ k}\Omega.$$

CB input resistance:

$$R_{\text{in}} = \frac{2500}{101} = \boxed{24.75 \Omega}.$$

Emitter-to-output gain:

$$A_{v,e} = +0.04(3197.3) = \boxed{+127.89}.$$

Including the 50Ω source:

$$A_{v,s} = 127.89 \frac{24.75}{50 + 24.75} \approx \boxed{+42.35}.$$

Under the cascode assumptions in the question, the lower transistor produces $g_m v_b$, and the upper transistor transfers that current to the output:

$$\boxed{A_{v,\text{cascode}} \approx -g_m R'_C = -127.89.}$$

The cascode input is at the lower transistor's base, rather than at a CB emitter. Its transistor input resistance is therefore approximately r_π , excluding its bias network. The cascode is inverting.

Q27. Two-stage cascode

Stage 1 is loaded by stage 2:

$$R'_{C1} = 4.7 \text{ k}\Omega \parallel 5 \text{ k}\Omega = 2.4227 \text{ k}\Omega,$$

$$A_{v1} = -0.04(2422.7) = \boxed{-96.91}.$$

Stage 2 is loaded by the final load:

$$R'_{C2} = 3.3 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 2.4812 \text{ k}\Omega,$$

$$A_{v2} = -0.02(2481.2) = \boxed{-49.62}.$$

Source attenuation:

$$\frac{v_{\text{in},1}}{v_s} = \frac{10}{1 + 10} = \frac{10}{11}.$$

Overall gain:

$$A_{v,s} = \frac{10}{11}(-96.91)(-49.62) \approx \boxed{+4372}.$$

Two inversions give a noninverting midband output.

Unloaded stage-gain product:

$$(-0.04 \times 4700)(-0.02 \times 3300) = (-188)(-66) = \boxed{12408}.$$

This overestimates the actual gain because it ignores interstage loading, output loading, and input attenuation.

For a final-output peak limit of 2 V:

$$V_{s,m} \approx \frac{2}{4372} = \boxed{0.457 \text{ mV}}.$$

This assumes small-signal operation and the stated first-stage swing condition.

Q28. Low-frequency corners

Input coupling:

$$f_{\text{in}} = \frac{1}{2\pi(1 + 10) \text{ k}\Omega(1 \mu\text{F})} = \boxed{14.47 \text{ Hz}}.$$

Output coupling:

$$f_{\text{out}} = \frac{1}{2\pi(2 + 10) \text{ k}\Omega(1 \mu\text{F})} = \boxed{13.26 \text{ Hz}}.$$

Emitter-bypass estimate:

$$f_E = \frac{1}{2\pi(100 \Omega)(100 \mu\text{F})} = \boxed{15.92 \text{ Hz}}.$$

These corners are close; no clearly separated dominant low-frequency pole is present. Consequently, choosing 15.92 Hz as the amplifier's lower cutoff is not a reliable combined-response calculation.

The bypass response generally has a pole and a zero, so its complete contribution also depends on the gain with and without bypass. The given isolated-corner information does not determine the exact combined lower cutoff.

Increasing C_{in} tenfold lowers its individual corner to approximately 1.447 Hz. The other two effects remain.

Q29. High-frequency response

Midband base-to-collector gain:

$$A_v = -g_m R'_C = -0.04(2000) = \boxed{-80}.$$

Miller input capacitance:

$$C_{\text{in,HF}} = 20 + 2[1 - (-80)] = \boxed{182 \text{ pF}}.$$

Resistance seen by this input capacitance:

$$R_{\text{in,HF}} = 1 \text{ k}\Omega \parallel 50 \text{ k}\Omega \parallel 2.5 \text{ k}\Omega \approx 704.23 \Omega.$$

Input-pole estimate:

$$f_{\text{p,in}} = \frac{1}{2\pi(704.23)(182 \text{ pF})} \approx \boxed{1.24 \text{ MHz}}.$$

Output capacitance:

$$C_{\text{out,HF}} = 10 + 2 \left(1 - \frac{1}{-80} \right) = \boxed{12.025 \text{ pF}}.$$

Output-pole estimate:

$$f_{\text{p,out}} = \frac{1}{2\pi(2000)(12.025 \text{ pF})} \approx \boxed{6.62 \text{ MHz}}.$$

The input pole is likely dominant; the approximate upper cutoff is near 1.24 MHz, subject to the limitations of the Miller pole approximation.

A cascode reduces signal voltage variation at the lower transistor's collector, reducing its BC capacitance's input Miller multiplication.

Q30. Power-amplifier calculations**(a) Series-fed Class A**

At the ideal midpoint:

$$V_{CEQ} = 6 \text{ V}, \quad I_{CQ} = \frac{12 - 6}{100} = \boxed{60 \text{ mA}}.$$

$$P_{DC} = 12(0.06) = \boxed{0.72 \text{ W}}.$$

With $I_m = 60 \text{ mA}$ and $V_m = 6 \text{ V}$:

$$P_{AC, \max} = \frac{V_m I_m}{2} = \frac{6(0.06)}{2} = \boxed{0.18 \text{ W}}.$$

$$\boxed{\eta_{\max} = 0.18/0.72 = 25\%}.$$

(b) Transformer-coupled Class A

Required reflected collector load:

$$R'_L = \frac{V_m}{I_m} = \frac{12}{0.06} = \boxed{200 \Omega}.$$

$$P_{DC} = 12(0.06) = 0.72 \text{ W},$$

$$P_{\text{out}, \max} = \frac{12(0.06)}{2} = \boxed{0.36 \text{ W}}, \quad \boxed{\eta_{\max} = 50\%}.$$

(c) Class B with dual supplies

$$I_m = \frac{V_m}{R_L} = \frac{10}{8} = \boxed{1.25 \text{ A}}.$$

Each supply carries a half-wave sinusoidal current, whose average over a full cycle is

$$I_{DC, \text{each}} = \frac{I_m}{\pi} = \boxed{0.398 \text{ A}}.$$

Total DC input power:

$$P_{DC} = 2V_{CC} \frac{I_m}{\pi} = 2(12) \frac{1.25}{\pi} = \boxed{9.55 \text{ W}}.$$

Output power:

$$P_o = \frac{V_m^2}{2R_L} = \frac{100}{16} = \boxed{6.25 \text{ W}}.$$

$$\boxed{\eta = \frac{6.25}{9.55} \approx 65.45\%}.$$

This is below the ideal 78.5% maximum because the output peak is 10 V, below the 12 V rail magnitude.

(d) Interpretation

- Series-fed Class A loses substantial DC power in its collector resistor and transistor.
- Ideal transformer coupling avoids collector-load DC resistive loss and permits greater voltage swing.

- Class B draws signal-dependent supply current rather than a continuous Class A quiescent current.
- Ideal Class B has zero supply current at idle. Practical Class AB intentionally has a small idle current.